

Wireshark BGP lab

Version 1.0, 30-10-2019, by Ivo Nijhuis

Getting started



Wireshark, retrieved on 30-10-2019 from [wireshark.org](https://www.wireshark.org)

In this lab we'll have a closer look at the BGP routing protocol by using Wireshark. To participate in this lab, you need to have Wireshark installed on your computer. You probably already have, but if not: download and install from <https://www.wireshark.org/>.

To start the lab, download the sample capture file containing a BGP communication between two routers from DLO.

Open the capture file in Wireshark and select `bgp` as display filter.

BGP Message types

1. What BGP message types do you see in the capture? What is the purpose of each of the message types? Refer to the sources listed at the end of the document if you're looking for more information!

BGP Open message

2. Have a closer look at the two open messages in the capture. What is the identifier for the AS each router is in? Are they in the same, or in different Autonomous Systems?

BGP Update message

3. The capture contains three update messages, containing route updates. For each update message, answer the following:
 - The update gives information about a route to reach which subnet?
 - What is the advertised NEXT_HOP router to reach this subnet?
 - What Autonomous Systems are contained in the AS_PATH attribute?

4. Explain how BGP uses the AS_PATH attribute to detect and prevent routing loops.

5. Will a BGP router always choose the route with the shortest AS_PATH length?
Explain your answer.

Complete the lab

To wrap up the lab, make a single document containing your answers. Upload your document to DLO.

Sources

The following sources contain a good introduction to help you make sense of BGP messages:

- Fundamentals of Border Gateway Protocol (BGP) - Part 1:
<https://kwtrain.com/blog/bgp-pt1>
- BGP Messages:
<https://networklessons.com/bgp/bgp-messages>